

Application of CFD High Fidelity Modelling: Nektar++

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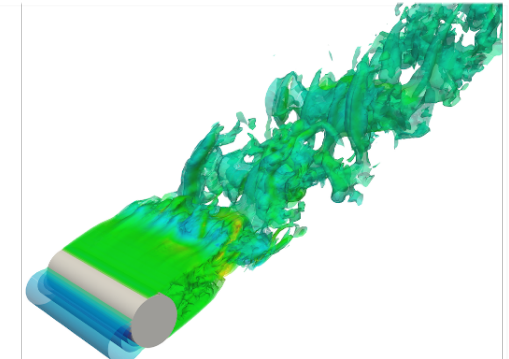
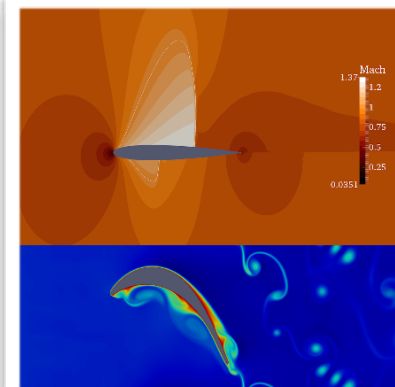
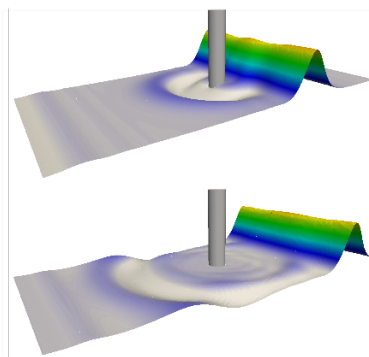
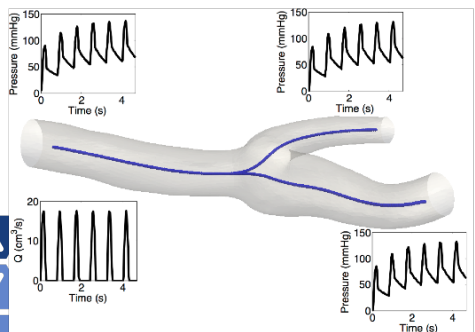
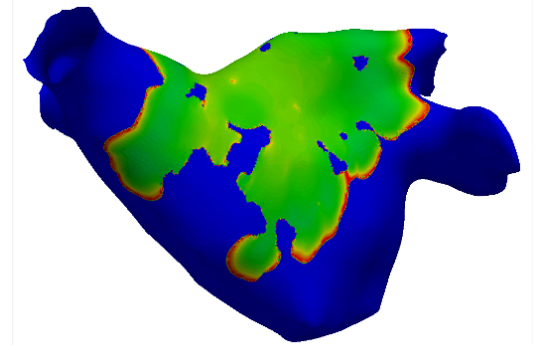
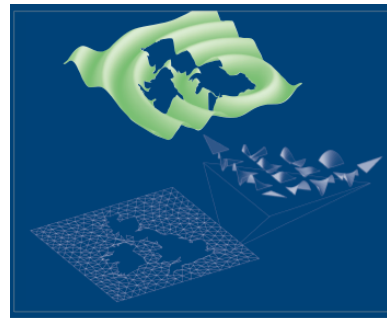
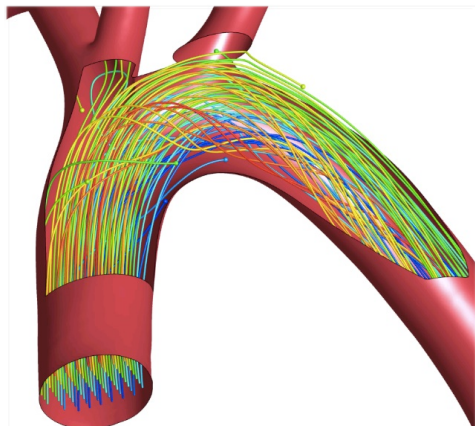
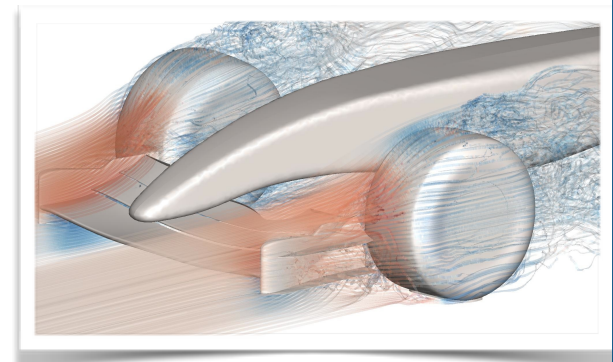
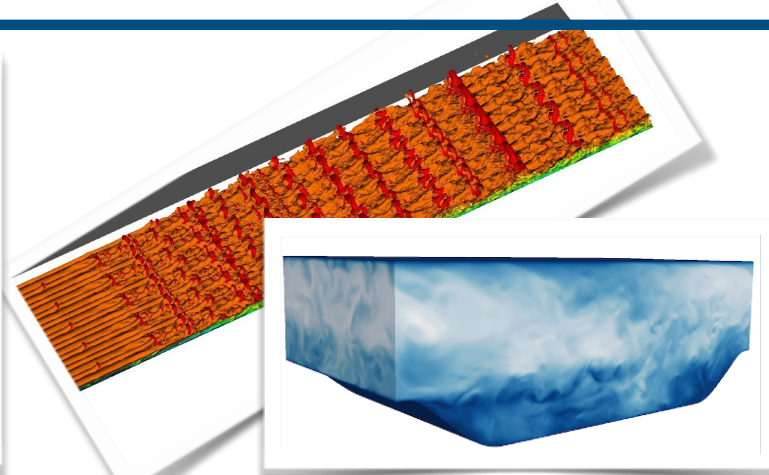
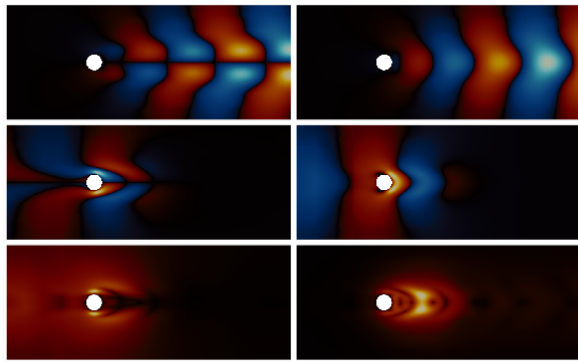


Flow Solvers





Nektar++: www.nektar.info



Incompressible Flow

- Navier-Stokes equation/Explicit time step
- Velocity Correction Scheme
- Pressure Boundary Conditions
- Example



Navier-Stokes Equations

$$\begin{aligned}
 \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} &= -\frac{1}{\rho} \frac{\partial p}{\partial x} + \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) \\
 \frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} &= -\frac{1}{\rho} \frac{\partial p}{\partial y} + \nu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right)
 \end{aligned}
 \quad \Rightarrow \quad
 \begin{aligned}
 \frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} &= -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u} \\
 \nabla \cdot \mathbf{u} &= 0
 \end{aligned}$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

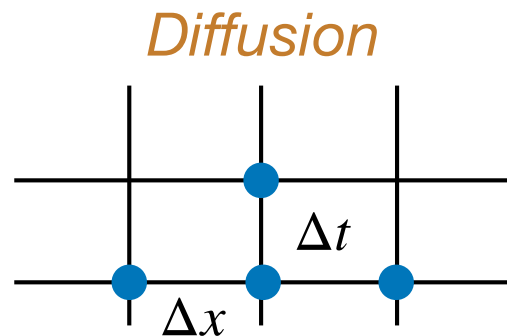
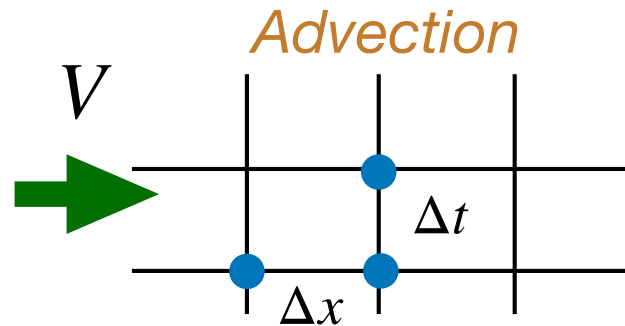
Advection

Diffusion

Spectral/hp element time step restrictions

Explicit Δt restrictions:

$$\frac{\partial u}{\partial t} = -V \frac{\partial u}{\partial x} + \sigma \frac{\partial^2 u}{\partial x^2}$$



Finite Difference

$$\Delta t \leq C \frac{\Delta x}{V}$$

$$\Delta t \leq C \frac{\Delta x^2}{\sigma}$$

Spectral/hp Element

$$\Delta t \leq C \frac{\Delta x}{V p^2}$$

$$\Delta t \leq C \frac{\Delta x^2}{\sigma p^4}$$

$$\Delta x \Rightarrow \frac{\Delta x}{p^2}$$



Navier-Stokes Solver

Navier-Stokes:
$$\partial_t \mathbf{u} + \mathbf{N}(\mathbf{u}) = -\nabla p + \nu \nabla^2 \mathbf{u}$$
$$\nabla \cdot \mathbf{u} = 0$$

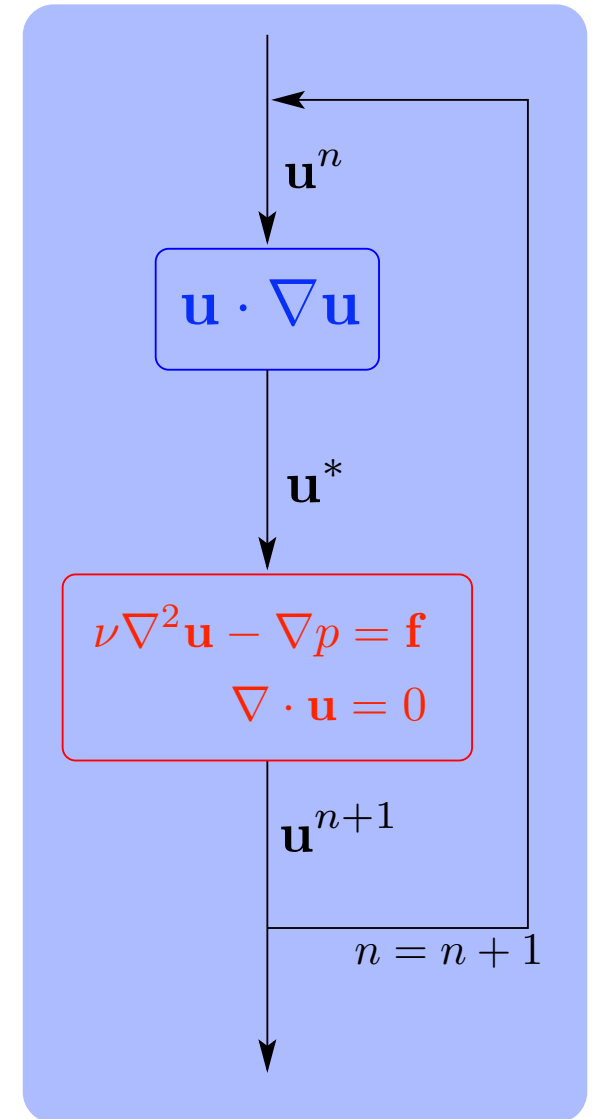
Velocity correction scheme (*aka stiffly stable*):

Orszag, Israeli, Deville (90), Karniadakis Israeli, Orszag (1991), Guermond & Shen (2003)

Advection:
$$\mathbf{u}^+ = -\sum_{q=1}^J \alpha_q \mathbf{u}^{n-q} - \Delta t \sum_{q=0}^{J-1} \beta_q \mathbf{N}(\mathbf{u}^{n-q})$$

Pressure Poisson:
$$\nabla^2 p^{n+1} = \frac{1}{\Delta t} \nabla \cdot \mathbf{u}^+$$

Helmholtz:
$$\nabla^2 \mathbf{u}^{n+1} - \frac{\alpha_0}{\nu \Delta t} \mathbf{u}^{n+1} = \frac{\mathbf{u}^+}{\nu \Delta t} + \frac{1}{\nu} \nabla p^{n+1}$$



Where does it come from?

$$\frac{\partial \mathbf{u}^{n+1}}{\partial t} + \mathbf{N}(\mathbf{u})^{n+1} = -\nabla p^{n+1} + \nu \nabla^2 \mathbf{u}^{n+1}$$

Backward Difference Formula:

$$\frac{\partial \mathbf{u}^{n+1}}{\partial t} = \frac{\alpha_0 \mathbf{u}^{n+1} - \hat{\mathbf{u}}}{\Delta t} + O(\Delta t^J)$$
$$\alpha_0 = \begin{cases} 1, & \text{if } J = 1, \\ \frac{3}{2}, & \text{if } J = 2, \end{cases} \quad \hat{\mathbf{u}} = \begin{cases} \mathbf{u}^n, & \text{if } J = 1, \\ 2\mathbf{u}^n - \frac{1}{2}\mathbf{u}^{n-1}, & \text{if } J = 2. \end{cases}$$

Extrapolation

$$\mathbf{N}^{n+1} = \mathbf{N}^{*,n+1} + O(\Delta t^J)$$

$$\mathbf{N}^{*,n+1} = \begin{cases} \mathbf{N}^n & \text{if } J=1 \\ 2\mathbf{N}^n - \mathbf{N}^{n-1} & \text{if } J=2 \end{cases}$$

$$\frac{\alpha_0 \mathbf{u}^{n+1} - \hat{\mathbf{u}}}{\Delta t} + \mathbf{N}(\mathbf{u})^{*,n+1} = -\nabla p^{n+1} + \nu \nabla^2 \mathbf{u}^{n+1}$$

Where does it come from?

$$\frac{\alpha_0 \mathbf{u}^{n+1} - \hat{\mathbf{u}}}{\Delta t} + \mathbf{N}(\mathbf{u})^{*,n+1} = -\nabla p^{n+1} + \nu \nabla^2 \mathbf{u}^{n+1}$$

Splitting in time:

$$\frac{\mathbf{u}^+ - \hat{\mathbf{u}}}{\Delta t} = -\mathbf{N}(\mathbf{u})^{*,n+1} \Rightarrow \mathbf{u}^+ = -\sum_{q=1}^J \alpha_q \mathbf{u}^{n-q} - \Delta t \sum_{q=0}^{J-1} \beta_q \mathbf{N}(\mathbf{u}^{n-q})$$

$$\frac{\hat{\mathbf{u}}^+ - \mathbf{u}^+}{\Delta t} = -\nabla p^{n+1} \Rightarrow ? \quad \nabla^2 p^{n+1} = \frac{1}{\Delta t} \nabla \cdot \mathbf{u}^+$$

$$\frac{\alpha_0 \mathbf{u}^{n+1} - \hat{\mathbf{u}}^+}{\Delta t} = \nu \nabla^2 \mathbf{u}^{n+1} \Rightarrow \nabla^2 \mathbf{u}^{n+1} - \frac{\alpha_0}{\nu \Delta t} \mathbf{u}^{n+1} = \frac{\mathbf{u}^+}{\nu \Delta t} + \frac{1}{\nu} \nabla p^{n+1}$$

Where does it come from?

$$\frac{\hat{\mathbf{u}}^+ - \mathbf{u}^+}{\Delta t} = -\nabla p^{n+1} \quad \Rightarrow ? \quad \nabla^2 p^{n+1} = \frac{1}{\Delta t} \nabla \cdot \mathbf{u}^+$$

Taking Divergence of RHS form:

$$\frac{\hat{\mathbf{u}}^+ - \mathbf{u}^+}{\Delta t} = -\nabla p^{n+1} \quad \Rightarrow \quad \nabla^2 p^{n+1} = \frac{1}{\Delta t} \nabla \cdot \mathbf{u}^+ \\ \text{if} \quad \nabla \cdot \hat{\mathbf{u}}^+ = 0$$

Splitting error

$$\nabla \cdot \hat{\mathbf{u}}^+ = 0 \quad \text{instead of} \quad \nabla \cdot \mathbf{u}^{n+1} = 0$$

$$\|\mathbf{u} - \mathbf{u}^h\|_{l^2(L^2(\Omega)^d)} \leq c\delta t^2.$$

Where does it come from?

$$\nabla^2 p^{n+1} = \frac{1}{\Delta t} \nabla \cdot \mathbf{u}^+$$

What about BCs and enforcing divergence condition?

Dot product of momentum equation:

$$\frac{\partial p^{n+1}}{\partial n} = \nabla p^{n+1} \cdot \mathbf{n} = \left[\nu \nabla^2 \mathbf{u}^{n+1} - \frac{\partial \mathbf{u}^{n+1}}{\partial t} - \mathbf{N}(\mathbf{u})^{n+1} \right] \cdot \mathbf{n}$$

$$\mathbf{N}^{n+1} \Rightarrow \mathbf{N}^{*,n+1}$$

$$\nabla^2 \mathbf{u} = \nabla (\nabla \cdot \mathbf{u}) - \nabla \times \nabla \times \mathbf{u}$$

Apply extrapolation and enforce zero divergence condition

$$\frac{\partial p^{n+1}}{\partial n} = \nabla p^{n+1} \cdot \mathbf{n} = - \left[\nu \nabla \times \nabla \times \mathbf{u}^{*,n+1} + \frac{\partial \mathbf{u}^{n+1}}{\partial t} + \mathbf{N}(\mathbf{u})^{*,n+1} \right] \cdot \mathbf{n}$$

But why do we need BCs on pressure when full equations do not have them?

Where do BC's come from?

Following Guermond & Shen:

$$\frac{\partial \mathbf{u}^{n+1}}{\partial t} + \mathbf{N}(\mathbf{u})^{n+1} = -\nabla p^{n+1} + \nu \nabla^2 \mathbf{u}^{n+1}$$

Inner product WRT ∇q directly get weak Poisson Eqn:

$$(\nabla q, \frac{\partial \mathbf{u}}{\partial t}) + (\nabla q, \mathbf{N}) = -(\nabla q, \nabla p) + \nu(\nabla q, \nabla^2 \mathbf{u})$$

Apply Divergence Theorem:

Where do BC's come from?

Following Guermond & Shen:

$$\frac{\partial \mathbf{u}^{n+1}}{\partial t} + \mathbf{N}(\mathbf{u})^{n+1} = -\nabla p^{n+1} + \nu \nabla^2 \mathbf{u}^{n+1}$$

Inner product WRT ∇q directly get weak Poisson Eqn:

$$(\nabla q, \frac{\partial \mathbf{u}}{\partial t}) + (\nabla q, \mathbf{N}) = -(\nabla q, \nabla p) + \nu(\nabla q, \nabla^2 \mathbf{u})$$

Apply Divergence Theorem:

$$(\nabla q, \frac{\partial \mathbf{u}}{\partial t}) = - (q, \nabla \cdot \frac{\partial \mathbf{u}}{\partial t}) + \langle q \frac{\partial \mathbf{u}}{\partial t} \cdot \mathbf{n} \rangle$$

$$(\nabla q, \mathbf{N}) = - (q, \nabla \cdot \mathbf{N}) + \langle q \mathbf{N} \cdot \mathbf{n} \rangle$$

$$(\nabla q, \nabla p) = - (q, \nabla^2 p) + \langle q \nabla p \cdot \mathbf{n} \rangle$$

$$(\nabla q, \nabla^2 \mathbf{u}) = - (\nabla q, \nabla \times \nabla \times \mathbf{u})$$

$$= (q, \nabla \cdot \nabla \times \nabla \times \mathbf{u}) - \langle q, \nabla \times \nabla \times \mathbf{u} \cdot \mathbf{n} \rangle$$

Where do BC's come from?

Following Guermond & Shen:

$$(\nabla q, \frac{\partial \mathbf{u}}{\partial t} + \mathbf{N} = -\nabla p + \nu \nabla^2 \mathbf{u})$$

Is equivalent to:

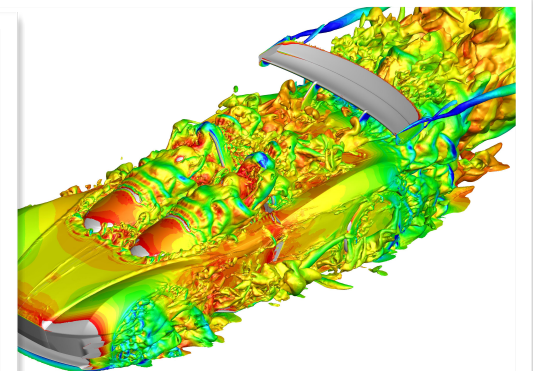
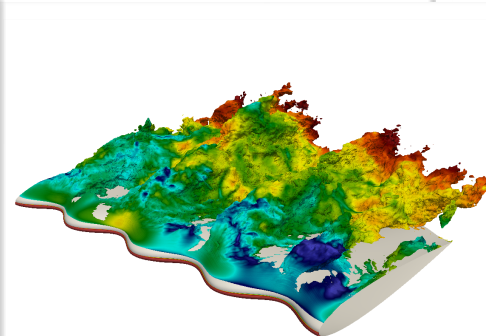
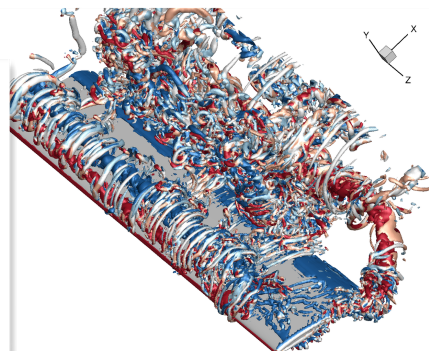
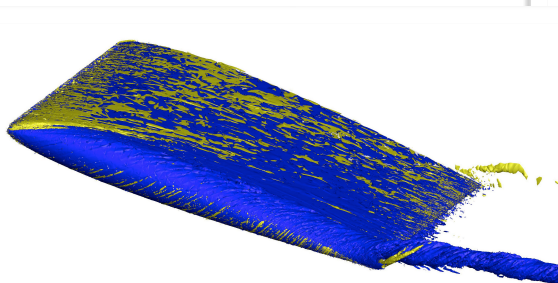
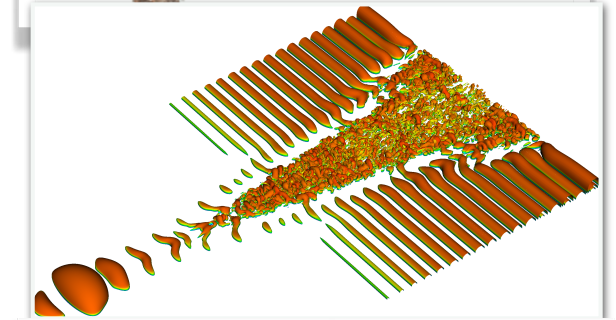
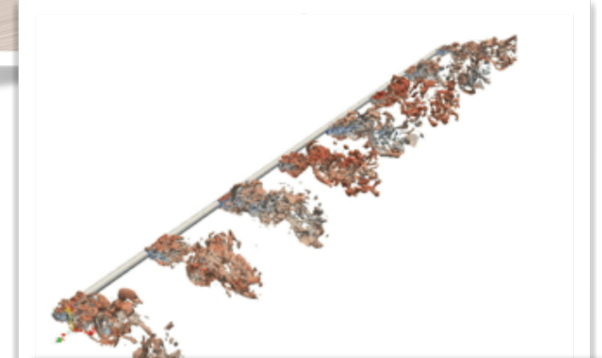
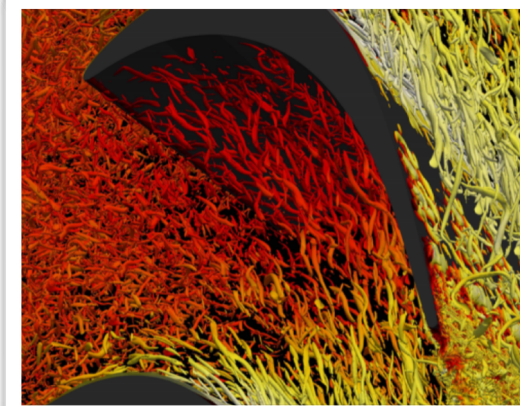
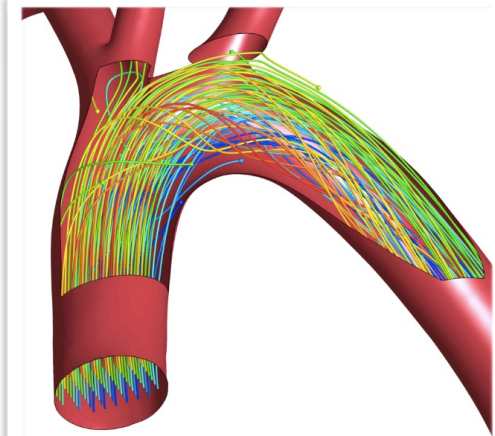
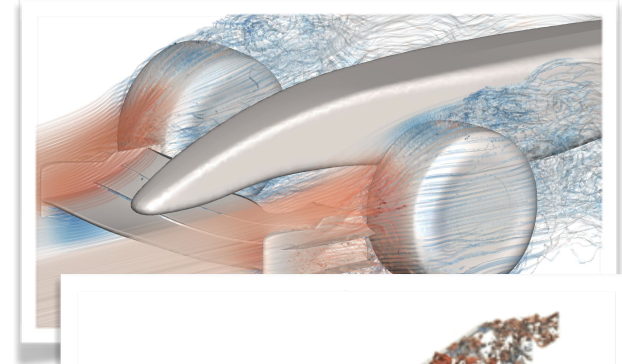
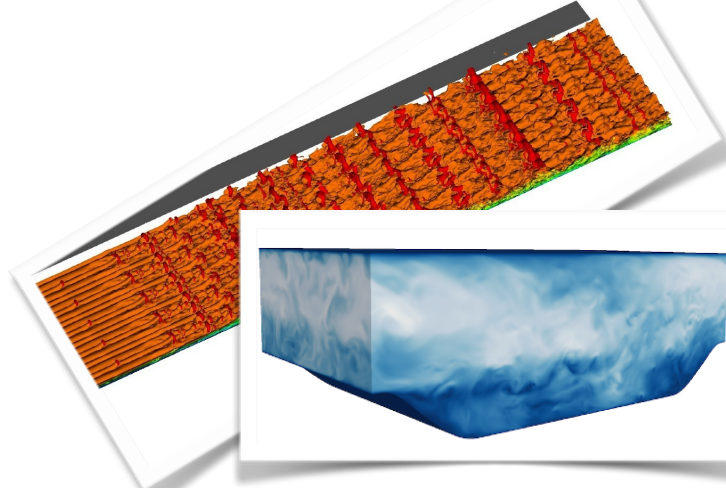
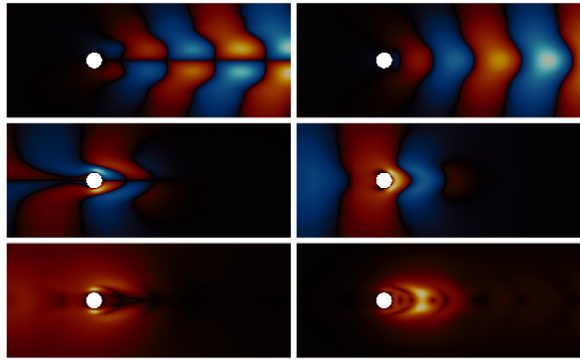
$$(q, \nabla \cdot \frac{\partial \mathbf{u}}{\partial t} - \nabla \cdot \mathbf{N} = \nabla^2 p)$$

$$\langle q, \frac{\partial \mathbf{u}}{\partial t} \cdot \mathbf{n} + \mathbf{N} \cdot \mathbf{n} = -\frac{\partial p}{\partial n} - \nu \nabla \times \nabla \times \mathbf{u} \cdot \mathbf{n} \rangle$$

$$\begin{aligned}
 (q, \nabla \cdot \frac{\partial \mathbf{u}}{\partial t} - \nabla \cdot \mathbf{N} = \nabla^2 p) &\Rightarrow \nabla^2 p^{n+1} = \frac{1}{\Delta t} \nabla \cdot \mathbf{u}^+ \\
 \langle q, \frac{\partial \mathbf{u}}{\partial t} \cdot \mathbf{n} + \mathbf{N} \cdot \mathbf{n} = -\frac{\partial p}{\partial n} - \nu \nabla \times \nabla \times \mathbf{u} \cdot \mathbf{n} \rangle &\Rightarrow \mathbf{u}^+ = -\sum_{q=1}^J \alpha_q \mathbf{u}^{n-q} - \Delta t \sum_{q=0}^{J-1} \beta_q \mathbf{N}(\mathbf{u}^{n-q}) \\
 &\Rightarrow \frac{\partial p^{n+1}}{\partial n} = - \left[\nu \nabla \times \nabla \times \mathbf{u}^{*,n+1} + \frac{\partial \mathbf{u}^{n+1}}{\partial t} \cdot \mathbf{n} + \mathbf{N}(\mathbf{u})^{*,n+1} \right] \cdot \mathbf{n}
 \end{aligned}$$

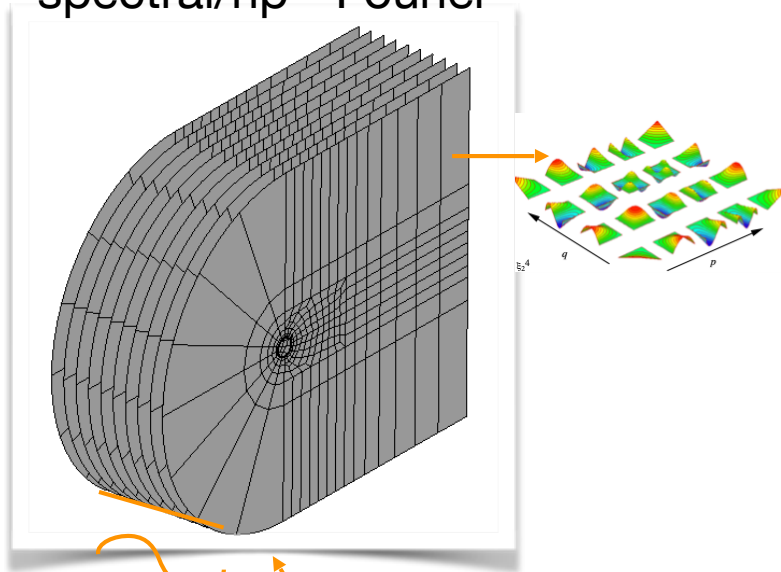


Nektar++: www.nektar.info

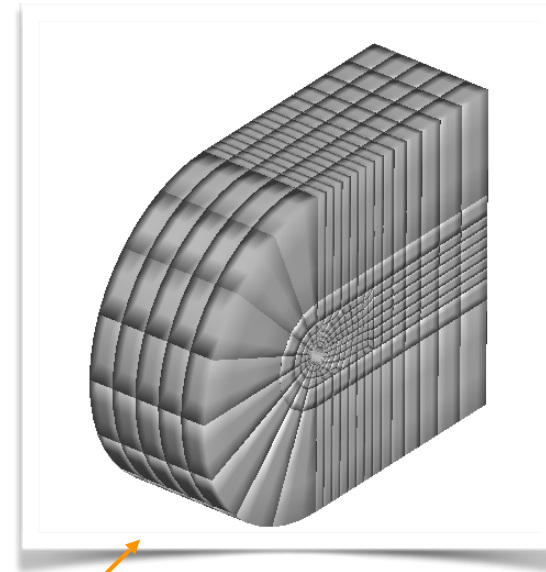


Discretisation Approaches

2.5D: Quasi 3D
spectral/hp - Fourier



Fully 3D



Direct

Iterative

Incompressible flow

$$\frac{\partial u}{\partial t} + u \cdot \nabla u = -\nabla p + \nu \nabla^2 u$$

IMEX:

$$\frac{\partial u}{\partial t} - \nu \nabla^2 u + \nabla p = -u \cdot \nabla u$$

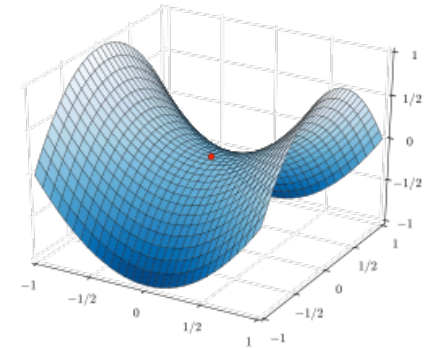
Implicit

explicit



Inf-Sup condition

- LBB condition/Inf-Sup
- Solving Stokes problem is so-called “saddle point problem”
- Pressure can be considered as a Lagrange multiplier that enforces divergence condition
- If pressure degrees of freedom are too many the problem can “lock”
- *Therefore often run with pressure polynomial order one lower than the velocity polynomial order*



Example: jupyter.ae.ic.ac.uk

/Tutorials/02-IncompressibleFlow/incns-solver.ipynb

1.1. Learning objectives

Welcome to the tutorial on the incompressible flow simulation using the Nektar++ framework. This tutorial aims to show how to setup and run a simulation for incompressible flows using the Nektar++ `IncNavierStokesSolver` for the well-known problem of external flow over a cylinder.

After completion of this tutorial you will be familiar with:

- Configuring a XML file suitable for incompressible Navier-Stokes in Nektar++.
- Simulating 2D and quasi-3D incompressible flows.
- Running the simulation in serial and parallel.
- Computing aerodynamic forces and modal energy as a function of time.
- Post-processing the results and converting the Nektar++ field outputs for visualization.

2. Introduction

In this tutorial we will walk through how to configure a session file to be able to solve incompressible Navier-Stokes equations using a simple flow example. We consider a circular cylinder. At the beginning we consider a two-dimensional case. Upon completion of the 2D case, we extend the problem to a quasi-3D case. By quasi-3D, a mesh will be extruded in the spanwise direction, with periodic boundary conditions applied in the spanwise direction.

2.1. Governing equation

The incompressible solver implemented in Nektar++ uses the convective form of the Navier-Stokes equations for viscous incompressible and Newtonian fluids, which can be written as:

$$\frac{\partial \mathbf{V}}{\partial t} + \mathbf{V} \cdot \nabla \mathbf{V} = -\nabla p + \nu \nabla^2 \mathbf{V} + \mathbf{f}$$
$$\nabla \cdot \mathbf{V} = 0$$

where $\mathbf{V} = (u, v, w)$ is the velocity of the fluid in the x , y , and z directions, respectively, p is the specific pressure (including density), ν is the kinematic viscosity, and $\mathbf{f}$ is a body force term.

3. Problem description: 2D case

In this tutorial we use air to simulate the flow past a 2D circular cylinder with diameter $D = 1$. For our study, we will use a Reynolds number $Re = 40$ with $Re = U \cdot D / \nu$, where U is the velocity scale, D is the length scale and ν is the kinematic viscosity.

The flow domain is a rectangle of sizes $[-5, 10] \times [-5, 5]$ as shown in the figure below.

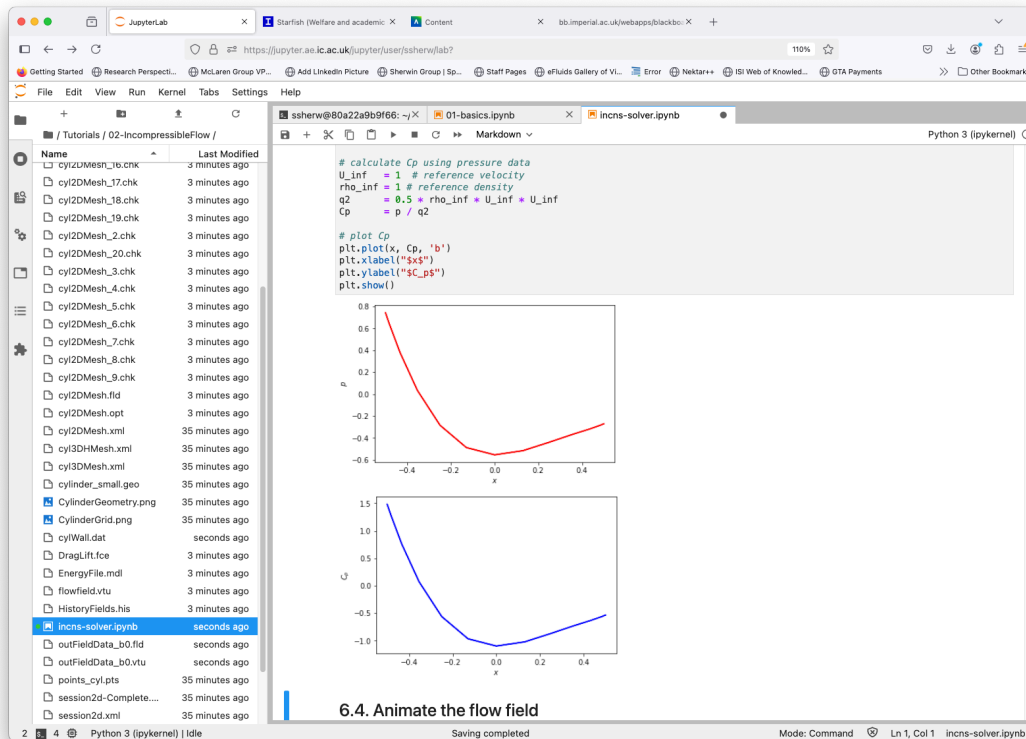
The figure shows a 2D plot of a cylinder in a flow field. The domain is a rectangle of size $[-5, 10] \times [-5, 5]$. The cylinder is centered at $(0, 0)$ with a diameter of $D = 1$. The flow is from left to right, with an inflow boundary at $x = -5$ and an outflow boundary at $x = 10$. The plot shows the velocity field around the cylinder, with a color scale ranging from -5 to 5.

Set up input file

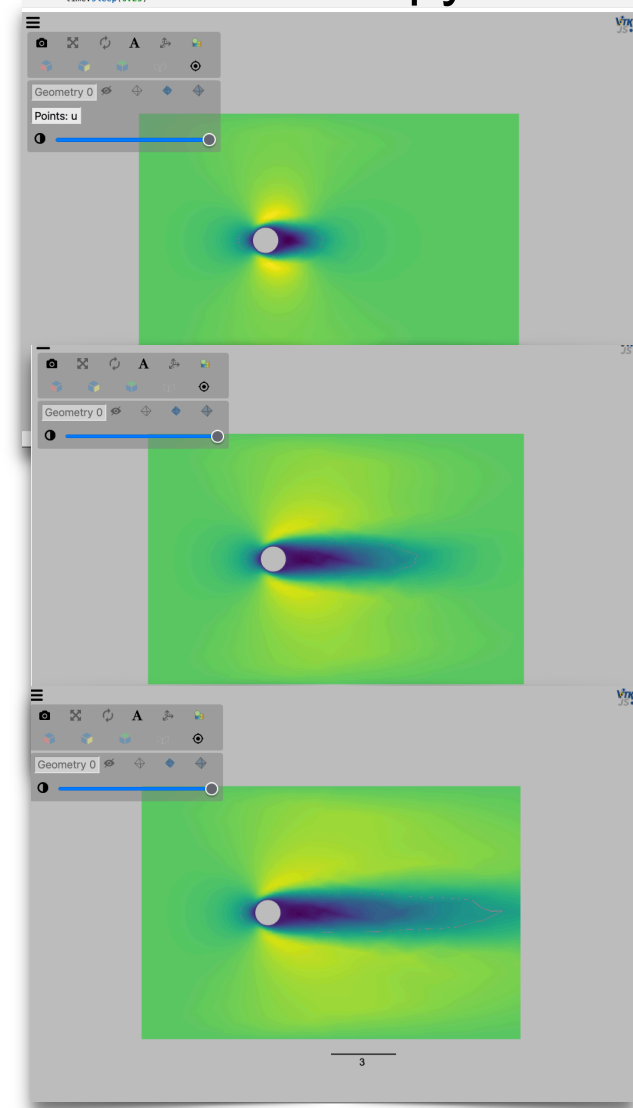
Run and plot the solution



Example: jupyter.ae.ic.ac.uk /Tutorials/02-IncompressibleFlow/incns-solver.ipynb



Extract the pressure distribution around cylinder



Animate the flow